

Erdős Problem #876

EP-876

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $A = \{a_1 < a_2 < \dots\} \subset \mathbb{N}$ be an infinite sum-free set - that is, there are no solutions to $a = b_1 + \dots + b_r$ with $b_1 < \dots < b_r < a \in A$. How small can $a_{n+1} - a_n$ be? Is it possible that $a_{n+1} - a_n < n$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>